

A Brief Review of End-to-End Visuomotor System

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I. DEEP VISUOMOTOR POLICIES WITH GPS

In this section, we will first review guided policy search [2] and then introduce the idea for end-to-end visuomotor in [3]. Policy search methods optimize a parameterized policy $\pi_\theta(u_t|x_t)$ over actions u_t conditioned on the state x_t . The goal is to minimize expected trajectory cost under the policy $J(\theta) = \sum_{t=1}^T \mathbb{E}_{\pi_\theta(x_t, u_t)}[J(c(x_t, u_t))]$ given cost $c(x_t, u_t)$ and dynamics $p(x_{t+1}|x_t, u_t)$. Denote the marginal of a trajectory $\tau = x_1, u_1, \dots, x_T$ as $\pi_\theta(\tau) = p(x_1) \prod_{t=1}^T p(x_{t+1}|x_t, u_t) \pi_\theta(u_t|x_t)$. The REINFORCE algorithm uses samples to compute $\nabla_\theta J(\theta)$ and often has large sample complexity. Instead guided policy search (GPS) methods splits the optimization into a “control (C) step” that finds multiple simple local policies $p_i(u_t|x_t)$ that for different initial states and a “supervised (S) step” that optimizes the global policy $\pi_\theta(u_t|x_t)$ to match all of the local policies using supervised learning [5]. These two steps respectively correspond to “E” and “M” steps for EM algorithm in machine learning. Effectively, a complex policy that takes raw observation “interpolates” the local controllers that are trained with state information. The benefit is that it requires substantially fewer samples due to the simplicity of the local policies. In [3], a policy with nearly a hundred thousand parameters for vision-based control can be trained under 200 episodes.

Note that the third step in algorithm 1 is critical because supervised learning might not guarantee that $\pi_\theta(u_t|x_t)$ will achieve long term performance as $p_i(u_t|x_t)$. Typically this is formulated as a constrained optimization that seeks at convergence to enforce equality:

$$\begin{aligned} \min_{\theta, p_1, \dots, p_N} \quad & \sum_{i=1}^N \sum_{t=1}^T \mathbb{E}_{p_i(x_t, u_t)}[c(x_t, u_t)] \\ \text{s.t. } \quad & p_i(x_t, u_t) = \pi_\theta(u_t|x_t) \quad \forall x_t, u_t, t, i \end{aligned} \quad (1)$$

Algorithm 1: Generic Guided Policy Search [5]

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for  $k = 0, \dots, K$  do
  C-step: improve each  $p_i(u_t|x_t)$  based on surrogate
    cost  $\hat{c}_i(x_t|u_t)$ , return samples  $\mathcal{D}_i$ 
  S-step: train  $\pi_\theta(u_t|x_t)$  with supervised learning on
    the dataset  $\cup_i \mathcal{D}_i$ 
  Modify  $\hat{c}_i(x_t|u_t)$  to enforce agreement between
     $\pi_\theta(u_t|x_t)$  and each  $p_i(u_t|x_t)$ 
end

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In practice, the local controller such iLQR collects samples to fit local linear-Gaussian dynamics for each local policy of the form $p_i(x_{t+1}|u_t, x_t) = \mathcal{N}(f_{x_t}x_t + f_{u_t}u_t + f_{c_t}, F_t)$ using linear regression. Then the controllers are improved in a modified LQR update with trust-region constraints $D_{KL}(\bar{p}_i(u_t|x_t)||p_i(u_t|x_t)) \leq \epsilon$. The surrogate objective $\hat{c}_i(x_t|u_t)$ similarly includes a term to encourage the local policy to be close to the global policy using $D_{KL}(p_i(u_t|x_t)||\pi_\theta(u_t|x_t))$. Specifically, Bregman ADMM (BADMM) is used in [3] to enforce the constraint $p_i(x_t, u_t) = \pi_\theta(u_t|x_t)$ at convergence and the improvement guarantees come from the algorithm. Minimizing the KL-divergence between the policy and trajectory distribution for conditional Gaussian policy is defined as $L_\theta(\theta, p)$. Intuitively it penalizes deviation from the trajectory distribution, with a penalty that is locally proportional to its cost-to-go. This constrained optimization approach can also be seen as a variant of mirror descent on the space of trajectory distributions where it first finds local controller that minimizes the cost and is close to the previous policy π_k , and then project the distribution to the parameter space [5]. This is related to other works in imitation learning such as DAGGER [6] except that the teacher (local controller) is now adapted to train the learner (global policy).

A few practical improvements are proposed in [3] to achieve efficient and robust training.. A neural network architecture is used to compute pixel-wise features points from raw observations. It turns out that the learned policy finds features on the target object and the robot gripper and arm. The network is pretrained for pose regression tasks in a self-supervised fashion. The algorithm is then used for simulation tasks including manipulation, walking, and swimming and real tasks involving tasks such as hanging a coat hanger and screwing on a bottle cap with a PR2 robot In both simulated and real experiments, the rewards are shaped to encourage reaching specified end effector poses. Specifically, in robot experiments, the cost is given by $c(x_t, u_t) = w_1 d_t^2 + w_2 \log(d_t^2 + \alpha) + w_3 \|u_t\|_2$ where w_1, w_2, w_3 are scalar weights. For each episode, the controllers have to reset to the exact start configuration for training purposes. In total, it takes around ten minutes for each task to be learned from scratch with high success rate. The ablation study shows that learning a visuomotor policy jointly finds more useful features and more likely to success than training each component separately. We will refer interested readers to the original paper [3] and follow up works [1, 2, 4, 5] for more technical details.

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